

Summary

Innovative software engineer with deep expertise in systems and graphics programming, medical imaging, and AI-driven solutions. Passionate about designing and implementing high-performance, scalable software architectures that bridge cutting-edge technology with healthcare innovation.

Education

- 2016–2023 **M.Sc. in Computer Science - Biomedical Computing**, *Technical University of Munich*, Munich, Germany
Coursework completed; thesis incomplete.
- 2010–2015 **B.Sc. in Electrical Engineering - Bioelectronics**, *Amirkabir University of Technology*, Tehran, Iran
Thesis: Smart MRA Injection Robot.
- 2009–2010 **Pre-University Diploma**, *Madani High School – National Organization of Exceptional Talents (NODET)*, Tabriz, Iran
- 2006–2009 **High School Diploma**, *Madani High School – National Organization of Exceptional Talents (NODET)*, Tabriz, Iran

Work Experience

- 07/2023–10/2024 **Senior Software Engineer**, *Holo-Light GmbH*, Munich, Germany
- Led XR system design with an emphasis on low-latency streaming.
 - Designed and implemented a cross-platform OpenXR library.
 - Developed Android XR applications using NDK, Java/Kotlin, and MediaCodec.
 - Created immersive 3D GUIs in Unity for AR/VR experiences.
 - Developed a WebRTC-based real-time communication service.
 - Implemented gRPC-based protocols for efficient microservice communication.
 - Optimized XR graphics using OpenGL ES, Vulkan, and GLSL.
 - Mentored team members on C++, algorithms, and clean code practices.
- 08/2022–03/2023 **Working Student – Software Developer**, *One-Projects GmbH*, Munich, Germany
- Engaged in C++ systems programming and object-oriented Python library development.
 - Developed applications using Qt and ImGui.
 - Enhanced build systems with Conan and Python scripts.
 - Implemented server load control scripts in Python.
- 02/2022–04/2022 **Working Student – Software Developer**, *AST (Adaptive Sensory Technology) GmbH*, Munich, Germany
- Performed C++ systems programming and graphic shader programming (GLSL/HLSL).
 - Set up cross-compilation pipelines using LLVM and CMake.
 - Configured remote access network infrastructure.
- 04/2021–06/2021 **Working Student – Software Developer**, *Fireflow GmbH*, Munich, Germany
- Developed iOS applications using SwiftUI.
 - Refactored and documented legacy iOS code.
 - Improved build systems and addressed user-reported issues.

- 08/2020–02/2021 **Working Student – Software Developer**, *SurgeVision GmbH*, Munich, Germany
- Developed algorithms for medical image processing.
 - Utilized Qt for application development and implemented test-driven computer vision algorithms (OpenCV, GoogleTest).
 - Set up Jenkins-based CI/CD pipelines.
- 08/2019–03/2020 **Working Student – Software Developer**, *Brainlab GmbH*, Munich, Germany
- Conducted research and developed algorithms for medical image processing using C++ and Python.
 - Developed computer vision algorithms, performed computational modeling, and created data visualizations.

Publications

- 2016 *Design and fabrication of a nanofibrous polycaprolactone tubular nerve guide for peripheral nerve tissue engineering using a two-pole electrospinning system* [Link]
- 2020 *Orientation of directional deep brain stimulation leads on CT: Resolving the ambiguity* [Link]

Skills

Programming C++, C#, Java, Kotlin, Python, Rust, Swift

Graphics & XR OpenGL, OpenGL ES, Vulkan, GLSL, OpenXR, Unity, Unreal Engine

Networking WebRTC, TCP/UDP, gRPC

Development Systems programming, cross-platform library design, CI/CD, build system optimization

Languages

English Proficient (C2)

German Intermediate (B1)

Others Persian, Azerbaijani, Turkish (Native)

Interests

Numerical Programming, Algorithms, Neuroscience, Computer Vision, Machine Learning, Computer Graphics, Software Engineering